14.3 Discovering Inverse Operations

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I can solve addition and subtraction one-step equations in one variable using manipulatives, drawings, and number lines to show the inverse operations.</td></tr>
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How would models help you solve an algebraic equation using inverse operations?</td></tr>
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Addition Property of Equality
Subtraction Property of Equality
<u>Additional Terms:</u>
Opposite</td><td>(Optional) Algebra tiles
(Optional) Blank tape diagrams
(Optional) Balance scale
(Optional) Marbles</td><td>N/A</td></tr>
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Teacher Tips

Common Misconceptions	Things to Consider
• Even though “opposites” and “inverse” can be used interchangeably in this lesson, be cautious that students will associate “inverse operations” with “opposites” in all cases. • Students may not have much experience with variables and be confused by a letter standing in for a number. • Students may struggle with integer operations. • Students may struggle to understand that what happens on one side of an equation must also occur on the other side to keep the equation balanced.	• Emphasize the visual of a balance beam or a hanger to help students understand and remember that meaning of the equals sign and that an operation that is done to one side of the equals sign of an equation must also be done on the other side of the equal sign. • Consistently switch the order of “addition” and “subtraction” in discussion to instill that both operations are important and relate to one another. • Tell students that while “opposites” and “inverse” can be used interchangeably in this lesson, that is not always the case. In mathematics, “opposites” refer specifically to two numbers whose sum is zero. • Highlight opportunities to clarify the distinction between “inverse operations” and “opposite values.” For example, in question 3 in 14.3.4 Guided Instruction, for $x + 5 = 11$, the “$+5$” represents both the inverse and opposite. • If time is a concern, consider assigning 14.3.5 Your Turn! for homework.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Notice and Wonder 14.3.2 Exploration positions students to use algebra tiles to represent and solve algebraic equations. Consider applying Notice and Wonder routines on each set of tiles, positioning students to discover the value of the variable through observation of the algebra tiles.	**MA.K12.MTR.6.1: Reasonableness** Students will be working to use inverse operations to solve one-step equations involving addition or subtraction. Clarifications within this standard emphasize the appropriateness of a solution to a problem. Encourage students to substitute their solution back into a problem and to verify their solution by explaining the method used.

Differentiate Instruction

Explain that the bar models, balance, hanger, and number line are all different ways of modeling the same concept of balancing an equation. Consider allowing students to use whichever method makes the most sense to them, like an actual hanger or balance to model the concept of balancing an equation by adding to or removing objects from each side of it. Allow students to use visual aids to solve equations for as long as they need to understand the concept. Encourage students to explain the process they use to solve the equations.

Lesson Notes:

14.3.1 Warm-Up: Picture This

Component Overview: *Students draw a visual representation of a mathematical situation and then use the image to solve the problem. Encourage students to explain the reasoning they used to solve the problem.*

14.3.1 Warm-Up: Picture This

Dale is helping to fix the roof at his uncle's house. They are using a 40-foot ladder. Dale is climbing the ladder to reach the roof. He is currently 7 feet up the ladder.

1. Sketch a picture to represent this situation.

   Answers will vary.

2. How much farther does Dale have to go to reach the roof? Include units.

   33 *feet*

After students complete the Warm-Up, have them explain how they arrived at their answer. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as:

- *"I drew…"*
- *"I figured out…"*

14.3.2 Exploration: Solving Equations Using Algebra Tiles

Component Overview: *Students use algebra tiles to understand the process of using inverse operations to solve algebraic equations. Students compare the number of algebra tiles on each side of an equation to determine the value of the variable tile. Students then discuss the equivalent relationship with a partner.*

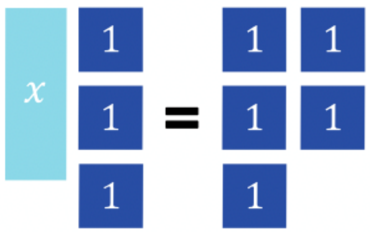
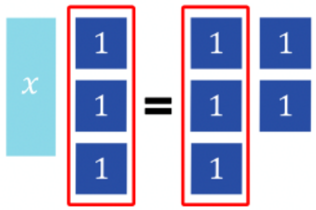
14.3.2 Exploration: Solving Equations Using Algebra Tiles

Algebra tiles can be used to represent and solve algebraic equations.

In the model shown, 1 represents 1 and x represents x.

1. The steps to determine the value of x using algebra tiles are shown.

   a. Work with your partner to complete the last step according to the description.

Algebra Tiles	Description
	Use algebra tiles to represent the equation $x + 3 = 5$.
	Determine any tiles that are represented on both sides of the equation.
	Remove the equal parts from both sides of the equation.

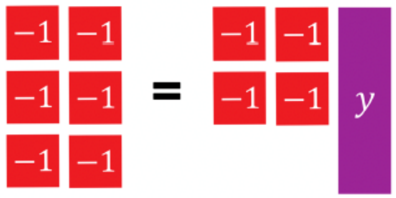
Consider allowing students to use algebra tiles to recreate the problem. Prompt students to discuss their rationale and approach as they work to continue building mathematical literacy.

b. The expressions $x + 3$ and 5 are equivalent according to the given equation. Discuss with your partner whether the tiles you drew on each side of the equation mat in the last step represent an equivalent relationship. Then, write a brief summary of your discussion.
Answers will vary. The tiles I drew represent an equivalent relationship because the original problem $x + 3 = 5$ represented an equivalent relationship, and since I removed the same number of tiles from each side of the equation, the relationship will still be equivalent.

c. How many blue tiles are equivalent to one teal tile?
2 blue tiles are equivalent to one teal tile.

d. What is the value of x in the equation $x + 3 = 5$?
$x = 2$

2. In the model shown, -1 represents -1 and y represents y.



a. Work with your partner to determine how many red tiles are equivalent to one purple tile.
2 red tiles are equivalent to one purple tile.

b. What is the value of y in the equation $-6 = -4 + y$?
$-2 = y$

Consider giving students a blank tape diagram. Have students use the diagram to build a model of the equation. Encourage students to explain how and why they are using the pictured algebra tiles to create a bar diagram.

Prompt students to make connections between their visual models and the learning targets for this lesson.
- *What do you notice about the equation?*
- *How does the equation relate to the tiles?*

14.3.3 Guided Instruction: Solving Equations Using Inverse Operations

Component Overview: *Students are introduced to the concept of using inverse operations to solve one-step equations. Students begin exploring the inverse operations of addition and subtraction.*

14.3.3 Guided Instruction: Solving Equations Using Inverse Operations

An equation relates two equivalent expressions. When solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

1. Discuss with your partner what it means for an equation to remain balanced. Then, summarize your discussion.

   Answers will vary. For an equation to remain balanced, you must add or subtract exactly the same quantity from both the left and the right side of the equation.

When finding the solution to an equation, inverse operations are used to balance an equation.

An ***inverse operation*** is an operation that reverses the effect of another operation.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	Inverse Operation
Subtraction $(-)$	*Addition $(+)$*
Addition $(+)$	*Subtraction $(-)$*

Work with students to define inverse operations. Discuss how the word "inverse" can be used in other contexts. Review the relationship of addition to subtraction to help students understand why inverse operations work when solving an equation.

Consider showing students inverse operations using arithmetic operations. Empathize the role of the equal sign.

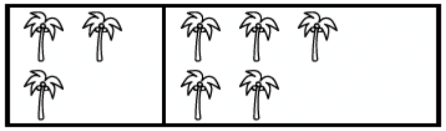
14.3.4 Guided Instruction: Solving Equations With Inverse Operations

Component Overview: *Students will begin exploring how to use inverse operations to solve one-step equations involving addition or subtraction. Students will use bar models and a balance scale to explore balancing on each side of an equation.*

14.3.4 Guided Instruction: Solving Equations with Inverse Operations

Bar models and balance scales are useful tools for representing equations.

1. In the bar model shown, each palm tree represents 1 unit.



The bar model can be used to represent either an addition or subtraction equation. Fill in the missing values in each equation to match the bar model.

$3 + \underline{\ 5\ } = 8$

$8 - 3 = \underline{\ 5\ }$

2. The bar model shown can be used to represent either an addition or subtraction equation.

7	
x	4

a. Circle the addition equation that matches the diagram.

$x + 7 = 4$ $7 = x + 4$ $x = 7 + 4$

b. Write an equivalent equation using the inverse operation.

Answers will vary. $x = 7 - 4$; $4 = 7 - x$

c. Justify to your partner that the equation you wrote is an equivalent equation.

Answers will vary. Since I used inverse operations to rewrite the equation, my equation is an equivalent equation because the value of the expressions on both sides of the equation remain equivalent.

After students create the equations, have them discuss how the equations relate to each other. Guide students toward the idea of addition and subtraction being inverse operations and what that means.

To provide additional scaffolding for students who may be struggling to make connections and identify the corresponding equations, ask questions like:

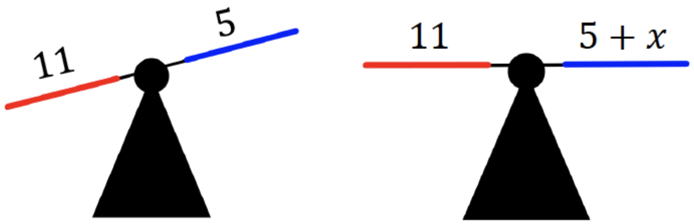
- *What do you notice about the bar model? What are the parts, and what represents the whole?*
- *How does the equation represent the bar model?*

Students could write $7 - 4 = x$ *OR* $7 - x = 4$. *Both would be correct, but one makes finding the value of* x *easier. Be ready to make those connections and listen for student thinking.*

14.3.4 Guided Instruction: Solving Equations With Inverse Operations (continued)

d. What is the value of the variable, x?
$x = 3$

3. Two balance scales are shown. The scale on the left is unbalanced. To balance the scale, an unknown value, x, needs to be added to the right side of the scale.



a. The equation represented by the balance scale on the right is shown. Use inverse operations to find the value of x by completing the steps shown.

$x + 5 = 11$

$x + 5 - \boxed{5} = 11 - \boxed{5}$

$x = \boxed{6}$

b. Rewrite the original equation, substituting the solution for x.

$5 + 6 = 11$

c. Explain how to use the equation in Part B to confirm that your solution is correct.
Answers will vary. The equation in Part B confirms that my solution is correct because the value of the expressions on both sides of the equal sign remain equivalent with my solution.

As the class works through question 3, use the following prompts to guide students toward making the connection between the visual of the balanced scales and balancing equations. This will build conceptual understanding that is foundational for lessons moving forward.

- *What do you notice about the balance scales? What is the same? What is different?*
- *How is the second balance scale related to the equation in question 3a?*

Inverse operations can be used to find the value of the variable in an algebraic equation by creating equivalent equations.

Addition Property of Equality

If $a = b$, then $a + c = b + c$

Subtraction Property of Equality

If $a = b$, then $a - c = b - c$

4. The equation $x + 23 = 78$ can be solved by using inverse operations.

   a. Fill in values in each equivalent equation to solve for x.

   $x + 23 = 78$
   $x + 23 - 23 = 78 - 23$
   $x = 55$

   b. What is the inverse operation used to solve the equation $x + 23 = 78$?
   Subtraction

   c. With your partner, discuss how to confirm whether or not the solution is accurate.
   Answers will vary. To confirm that the solution is accurate, I can substitute my solution into the original equation and confirm that the values on each side of the equal sign are equivalent to each other.

Refer to MA.K12.MTR.6.1 as you remind students to assess whether their solution is reasonable and to substitute their answer back into the equation as a way to verify their solution.

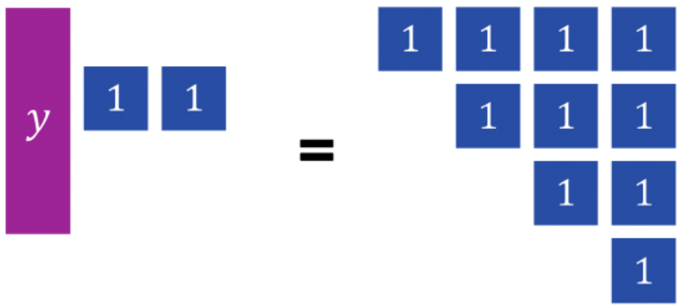
Have students construct an equivalent equation for $x + 23 = 78$ and explain how and why the equations are equivalent. Also encourage students to construct another pair of equivalent equations using addition or subtraction and then explain how inverse operations can be used to solve them.

14.3.5 Your Turn!

Component Overview: *Students will practice using the inverse operations of addition and subtraction to solve one-step equations.*

14.3.5 Your Turn!

1. The algebra tiles below model an equation.



   a. How many blue tiles are equivalent to one purple tile?
   8 *blue tiles are equivalent to one purple tile.*

   b. Write a one-variable equation to represent the tiles on the mat.
   $y + 2 = 10$

   c. What inverse operation can be used to solve the equation?
   Subtraction

   d. What value of y makes the equation true?
   $y = 8$

2. The equation $-31 = s - 6$ can be solved using inverse operations.
   a. Fill in the missing information to solve for s.

$$-31 = s - 6$$
$$-31 + 6 = s - 6 + 6$$
$$-25 = s$$

Consider an inquiry-based learning approach to facilitate connections between visual models and abstract concepts.

- *How can you figure out the relationship between the blue tiles and purple tiles?*
- *How can you use this relationship to solve the equation?*
- *Why do you use inverse operations to solve the equation?*

b. Which inverse operation was used to solve the equation?
Addition

c. Check your solution.
$-31 = -25 - 6$
$-31 = -(25 + 6)$
$-31 = -31$

3. Two bar model diagrams are shown.

Bar Model Diagram 1	**Bar Model Diagram 2**
m	-12
-12 | -4	4 | m

a. Write an algebraic equation for each diagram box diagram.

Algebraic Equation for Diagram 1	**Algebraic Equation for Diagram 2**
Answers will vary. $-12 + (-4) = m$ $m = -4 + (-12)$	*Answers will vary.* $4 + m = -12$ $-12 = m + 4$

b. Find the value of variable, m, for each algebraic equation.

Diagram 1	**Diagram 2**
$-16 = m$	$m = -16$

c. Discuss with your partner what you notice about the two equations. Summarize your discussion.
Answers will vary. I noticed that that the solutions to both equations are the same even though the equations have different values.

Solving this question requires students to apply the inverse operation of addition to both sides of the equation. Reiterate the importance of assessing the reasonableness of a solution and substituting the solution back into the equation as a way to check it.

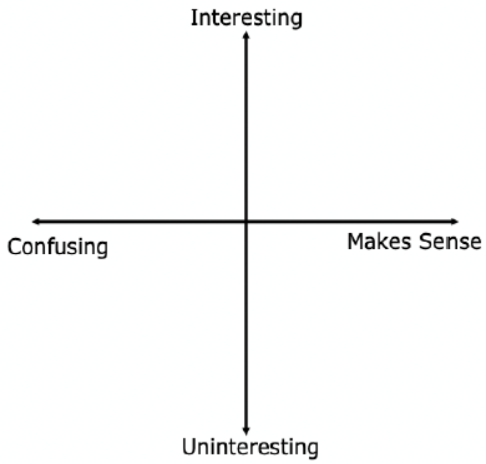
14.3.6 Wrap-Up: Reflection

Component Overview: *Students will reflect on their learning and understanding of the lesson. Encourage students to not only plot a point representing how they feel about meeting the learning target but to also explain why they feel that way.*

14.3.6 Wrap-Up: Reflection

1. Plot a point to indicate how you feel about today's learning target: "I can use diagrams and inverse operations when solving one-step addition and subtraction equations."

   Answers will vary.



Consider prompting students to include the learning target as an ordered pair and explaining why they chose that value. For example, a student may choose $(-1, 5)$ instead of $(-4, 5)$, explaining that it was confusing at first but made more sense by the end of the lesson.